Experiment-2

Aim:- To convert the ER diagram for an Indian e-commerce platform into a relational schema and implement it using MySQL CREATE TABLE statements with primary keys, foreign keys, NOT NULL, UNIQUE, ON DELETE CASCADE and SET NULL constraints.

Relational Schema

CUSTOMER- Customer_ID , Name, Email , Phone_No

SELLER- Seller_ID, Name, Email , Phone_No , GST_No

CATEGORY- Category_ID, Category_Name , Description

ADDRESS- Address_ID, House_No, Street, City, State Pincode , Customer_ID

PRODUCT- Product_ID , Product_Name, Price , Images , Description , Seller_ID , Category_ID.

ORDER- Order_ID , Order_Date , Status , Customer_ID

ORDER_ITEM- Order_ID , Product_ID , Quantity

PAYMENT- Payment_ID , Payment_Date , Amount, Payment_Mode , Order_ID

DELIVERY- Delivery_ID , Delivery_Date , Status, Courier_Name, Tracking_No , Order_ID, Address_ID.

Keys -

- Primary Keys:- Customer_ID, Seller_ID, Category_ID, Product_ID , Order_ID , Product_ID + Order_ID, Payment_ID, Delivery_ID , Address_ID.
- Foreign Keys:- Seller_ID, Category_ID , Customer_ID , Order_ID, Product_ID, Address_ID.

Source Code

```
CREATE DATABASE ECommerceDB;
USE ECommerceDB;

CREATE TABLE Customer (
    Customer_ID INT PRIMARY KEY,
    Name VARCHAR(100) NOT NULL,
    Email VARCHAR(100) NOT NULL UNIQUE,
    Phone_No VARCHAR(15) UNIQUE
);

CREATE TABLE Seller (
    Seller_ID INT PRIMARY KEY,
    Name VARCHAR(100) NOT NULL,
    Email VARCHAR(100) NOT NULL UNIQUE,
    Phone_No VARCHAR(15) UNIQUE,
    GST_No VARCHAR(20) UNIQUE,
);

CREATE TABLE CATEGORY (
    Category_ID INT PRIMARY KEY,
    Category_Name VARCHAR(50) NOT NULL UNIQUE,
    Description VARCHAR(200)
);

CREATE TABLE Address (
    Address_ID INT PRIMARY KEY,
    House_No VARCHAR(20) NOT NULL,
    Street VARCHAR(100),
    City VARCHAR(50) NOT NULL,
    State VARCHAR(50) NOT NULL,
    Pincode VARCHAR(50) NOT NULL,
    Customer_ID INT,
    FOREIGN KEY (Customer_ID)
        REFERENCES Customer(Customer_ID)
        ON DELETE SET NULL
);
```

```
CREATE TABLE Orders (
    Order_ID INT PRIMARY KEY,
    Order_Date DATE NOT NULL,
    Status VARCHAR(30) NOT NULL,
    Customer_ID INT NOT NULL,

    FOREIGN KEY (Customer_ID)
        REFERENCES Customer (customer_ID)
        ON DELETE CASCADE
);

CREATE TABLE Order_Item(
    Order_ID INT,
    Product_ID INT,
    Quantity INT NOT NULL,

    PRIMARY KEY (Order_ID, Product_ID),
    FOREIGN KEY (Order_ID)
        REFERENCES Orders (Order_ID)
        ON DELETE CASCADE
);

CREATE TABLE Delivery (
    Delivery_ID INT PRIMARY KEY,
    Delivery_Date DATE,
    Status VARCHAR(30) NOT NULL,
    Courier_Name VARCHAR(100),
    Tracking_No VARCHAR(50) UNIQUE,
    Order_ID INT NOT NULL UNIQUE,
    Address_ID INT,
    FOREIGN KEY (Order_ID)
        REFERENCES Orders (Order_ID)
        ON DELETE CASCADE,
    FOREIGN KEY (Address_ID)
        REFERENCES Address (Address_ID)
        ON DELETE SET NULL
```

Insert Sample Data

```
INSERT INTO Customer
VALUES
(1, 'Kavya Kushwaha', 'kavya@gmail.com', '9076543210'),
(2, 'Rahul Sharma', 'rahul@gmail.com', '9056743211');
INSERT INTO Seller
VALUES
(101, 'Tech Store', 'tech@gmail.com', '9076500011', 'GST12345'),
(102, 'Fashion Hub', 'fashion@gmail.com', '9076500022', 'GST123456'),
INSERT INTO Category
(201, 'Electronics', 'Electronic products'),
(202, 'Clothing', 'Clothing Products');

INSERT INTO Address
VALUES
(301, '12A', 'MG Road', 'Lucknow', 'Uttar pradesh', '226001', 1),
(302, '25B', 'Park Road', 'Delhi', 'Delhi', '110001', 2);
INSERT INTO Product
VALUES
(401, 'Laptop', 55000.00, 'laptop.jpg',
    'Gaming Laptop', 101, 201),
(402, 'T-Shirt', 999.00, 'tshirt.jpg',
    'Cotton T-Shirt', 102, 202);
INSERT INTO Order
VALUES
(501, '2026-08-19', 'Confirmed', 1),
(502, '2026-08-19', 'Pending', 2);
INSERT INTO Order_Item
VALUES
(501, 401, 1),
(502, 402, 2);
INSERT INTO Payment
Values
(601, '2026-08-19', 55000.00, 'UPI', 501),
(602, '2026-08-19', 1998.00, 'COD', 502);
```

```
INSERT INTO Delivery
VALUES
(701, '2026-08-22', 'Shipped',
   'BlueDart', 'TRK001', 501, 301),
(702, '2026-08-23', 'Pending',
   'Delhivery', 'TRK1002', 502, 302);
```

Referential Integrity Violations

1) Invalid Customer ID

```
INSERT INTO Orders VALUES
(503, '2026-08-19', 'Pending', 999);
```

Output - Foreign key constraint fails because CUSTOMER_ID=999 does not exist.

2) Invalid ProductID

```
INSERT INTO Order_Item Values
(501, 999, 1);
```

Output - Foreign key constraint fails because Product_ID=999 does not exist.

3) ON DELETE CASCADE

```
DELETE FROM Customer
WHERE Customer_ID=1;
```

Result - Related orders are automatically deleted.

4) ON DELETE SET NULL

```
DELETE FROM Seller
WHERE Seller_ID =101;
```

Output - Related Product.Seller_ID becomes NULL.

Output -

The relational schema was successfully implemented using Primary Key, Foreign Key, NOT NULL, UNIQUE ON DELETE CASCADE and ON DELETE SET NULL constraints. Sample data and referential integrity violations were successfully demonstrated.